

A Convolution Algebra of Stieltjes Functions

Finite-part identities, polygamma contact terms,
log-gamma convolution powers, and polylogarithmic correlations

Research continuation prepared for Vladimir Reshetnikov
ProveIt / Analysis / Polylogarithms

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Abstract

The Hurwitz-zeta convolution semigroup has a singular parameter at which its scalar formula ceases to describe the full periodic distribution. Keeping the missing contact term produces a closed convolution algebra for all generalized Stieltjes functions. We prove an all-index Bell-polynomial normal form, a bivariate Gamma-quotient multiplication law, and absolutely convergent subtracted-integral realizations away from the singular point. The first identity is a finite-part digamma convolution equal to $2\gamma_1(x) + \zeta(2)$. Differentiation gives an explicit anomaly: periodic differentiation and canonical Hadamard finite part differ by a known multiple of a derivative of the delta distribution. Consequently, every convolution of two finite-part polygamma functions reduces to just two Hurwitz-zeta jets, with corrected harmonic-number coefficients. At the integrated end, every circular convolution power of centered $\log \Gamma$ is a finite combination of jets at a single negative integer. Reflection changes the same spectral calculation into order derivatives of polylogarithms and yields a full shifted log-gamma correlation formula. All analytic statements are proved in the article. Exact finite algebra and independent high-precision numerical diagnostics are supplied separately. The basic nonsingular semigroup is attributed to prior work; no priority claim, period-independence claim, or proof-assistant verification is made.

Contents

1	Scope, provenance, and main conclusions	3
1.1	What is established here	3
1.2	Classical input and novelty boundaries	3
2	Fourier conventions and the Hurwitz semigroup	4
3	Canonical finite parts of the Stieltjes functions	5
4	All-index convolution closure and exact normal forms	6
5	Ordinary integral realizations and explicit Stieltjes identities	8
5.1	A concrete digamma integral	9
6	Differentiation and the contact-term correction	9
6.1	Relationship with the manuscript's master identity	10
6.2	Convolutions of arbitrary differentiated Stieltjes functions	11
7	A two-jet collapse for every polygamma convolution	11
7.1	A convergent trigamma identity with every subtraction visible	12
8	Antiderivatives and log-gamma convolution powers	13
9	Reflection, polylogarithmic order derivatives, and shifted correlations	14
10	Audit findings and proposed manuscript integration	16
11	Verification and reproducibility	17
11.1	Finite exact algebra	17
11.2	Independent numerical diagnostics	17
11.3	Rebuilding	18
12	Further research questions	18
12.1	Twists, characters, and alternate endpoint normalizations	18
12.2	Affine arguments and the boundary of depth-one reduction	18
12.3	Collision limits and renormalization compatibility	18
12.4	Cyclotomic integration certificates	18
12.5	Negative-integer jets and arithmetic reductions	19
12.6	Minimal formalization interfaces	19
13	Conclusion	19

1 Scope, provenance, and main conclusions

This continuation addresses identities rather than inequalities. Its starting point is the integrated and differentiated Hurwitz-jet calculus in the ProveIt manuscript [1], especially the chapters `07-integration.tex` and `08-differentiation.tex`. The repository snapshot used for the final source audit is

0efdbaa6944357e578a8839ec88d040c6e05681c.

The surviving-problems chapter in that snapshot already calls the S_4 identity proved. We do not present it as an open problem, nor claim a new solution of the remaining S_6 or arithmetic-independence questions. The incoming-directory inventory and intake instructions were inspected; the binary incoming archives were not individually extracted or exhaustively audited. These limits matter when comparing this contribution with a rapidly changing research repository. Detailed source paths and suggested placement are in the accompanying audit and integration notes.

1.1 What is established here

The principal addition is a normalization-aware identity calculus. If G_n is the canonical periodic finite part of γ_n and $l = \delta_0 - 1$, then

$$G_0 * G_0 = 2G_1 - \zeta(2)l. \quad (1.1)$$

The star denotes circular convolution, not pointwise multiplication. Restricting away from 0 (mod 1) changes the last term into $+\zeta(2)$. More generally, all $G_m * G_n$ are finite linear combinations of $l, G_0, \dots, G_{m+n+1}$, with explicit zeta-polynomial coefficients. There is no unresolved analytic remainder.

The derivative correction is equally explicit. Write $e_j(k) = e_j(1, 1/2, \dots, 1/k)$, setting $e_j(k) = 0$ for $j > k$. Then

$$D^k G_n = \text{Fp}_{\mathbb{T}}[\gamma_n^{(k)}] + (-1)^{n+1} n! e_{n+1}(k) \delta_0^{(k)} \quad (k \geq 1). \quad (1.2)$$

The ordinary pointwise parameter-derivative theorem in the repository remains valid. Equation (1.2) specifies the extra term needed when that theorem is extended to periodic finite parts.

For example, putting $\Psi_k = \text{Fp}_{\mathbb{T}} \psi^{(k)}$, one obtains, for $0 < x < 1$,

$$(\Psi_1 * \Psi_1)(x) = -4\zeta'(3, x) - 2\zeta(3, x). \quad (1.3)$$

A completely explicit, absolutely convergent subtracted-integral version appears in Section 7. Thus distribution notation is a tool for derivation, not an obstacle to ordinary numerical evaluation.

1.2 Classical input and novelty boundaries

Hurwitz's Fourier formula and the usual Gamma, polygamma, and polylogarithm identities are classical; convenient primary reference compilations are DLMF Sections 25.11 and 25.12 [3, 4]. The nonsingular Hurwitz convolution law below is already stated by Zhi-Hong Sun in Remark 3.2 of *On the properties of invariant functions* [2], for real parameters greater than one. We supply a short Fourier proof and continue it as an entire distribution-valued family. Espinosa

and Moll's integral calculus [5, 6] supplies important context at the integrated end; Coffey's all-order Stieltjes derivative formulas [7] supply context at the differentiated end.

The all-index finite-part multiplication table, explicit local subtractions, contact correction, polygamma collapse, and convolution-power formulas are the concrete results developed in this continuation. A literature search located the basic semigroup and the classical moment and derivative theories, but did not establish a comprehensive priority history for every specialization below. Accordingly, "proved here" is not used as a synonym for "first proved in the literature."

2 Fourier conventions and the Hurwitz semigroup

We work on $\mathbb{T} = \mathbb{R}/\mathbb{Z}$, with normalized Lebesgue measure. For a periodic distribution T , use the convention

$$\widehat{T}(k) = \langle T, e^{-2\pi i k x} \rangle, \quad k \in \mathbb{Z}.$$

The constant function 1 and the point mass δ_0 both have mean one, but their nonzero Fourier coefficients are respectively 0 and 1. Thus

$$\mathbf{l} = \delta_0 - 1$$

is the convolution identity on the space of mean-zero distributions. In particular, $\mathbf{l} * T = T$ whenever $\widehat{T}(0) = 0$.

The logarithm used throughout is

$$\lambda_k = \log(2\pi i k) = \log(2\pi |k|) + \frac{\pi i}{2} \operatorname{sgn} k, \quad k \neq 0. \quad (2.1)$$

All powers below refer to this fixed branch.

Definition 2.1. For $\alpha \in \mathbb{C}$, define R_α by

$$\widehat{R}_\alpha(0) = 0, \quad \widehat{R}_\alpha(k) = e^{-\alpha \lambda_k} \quad (k \neq 0). \quad (2.2)$$

Equivalently, its Fourier series is $\sum_{k \neq 0} (2\pi i k)^{-\alpha} e^{2\pi i k x}$ in the distribution sense.

Lemma 2.2 (Entire distribution family). *The family $\alpha \mapsto R_\alpha$ is entire as a periodic-distribution family. It is represented by an absolutely convergent Fourier series for $\operatorname{Re} \alpha > 1$. For $\operatorname{Re} \alpha > 0$, it is the locally integrable periodic extension of*

$$R_\alpha(x) = \frac{\zeta(1 - \alpha, x)}{\Gamma(\alpha)}, \quad 0 < x < 1. \quad (2.3)$$

Proof. On each compact set of parameters, the coefficients in (2.2), together with any fixed number of parameter derivatives, have at most polynomial growth in $|k|$. Fourier coefficients of a smooth test function decay faster than every polynomial. Hence the paired Fourier series and all its parameter derivatives converge locally uniformly. This proves the first assertion; the absolute-convergence assertion follows immediately from $\sum k^{-\operatorname{Re} \alpha}$.

For $\operatorname{Re} \alpha > 1$, equation (2.3) is precisely Hurwitz's Fourier formula. To continue to $\operatorname{Re} \alpha > 0$, use

$$\zeta(1 - \alpha, x) = x^{\alpha-1} + \zeta(1 - \alpha, 1 + x).$$

On compact parameter subsets of that half-plane, the first term and its parameter derivatives are dominated by integrable powers of x times powers of $|\log x|$. The second term is regular in $x \in [0, 1]$; any apparent parameter singularity at $\alpha = 0$ is outside the stated open half-plane. The two holomorphic distribution families agree in an open subregion, and hence agree throughout the half-plane. $\square$

Theorem 2.3 (Semigroup and differential ladder). *For all $\alpha, \beta \in \mathbb{C}$,*

$$R_\alpha * R_\beta = R_{\alpha+\beta}, \quad DR_\alpha = R_{\alpha-1}, \quad R_0 = 1. \quad (2.4)$$

If $J_{\alpha,r} = \partial_\alpha^r R_\alpha$, then

$$J_{\alpha,r} * J_{\beta,s} = J_{\alpha+\beta,r+s}, \quad D^k J_{\alpha,r} = J_{\alpha-k,r}. \quad (2.5)$$

Proof. Periodic distributions can be convolved: equivalently, their polynomially growing Fourier coefficients can be multiplied. For $k \neq 0$,

$$e^{-\alpha\lambda_k} e^{-\beta\lambda_k} = e^{-(\alpha+\beta)\lambda_k}, \quad (2\pi i k) e^{-\alpha\lambda_k} = e^{-(\alpha-1)\lambda_k}.$$

Both sides have zero zeroth coefficient. Fourier uniqueness proves (2.4). Differentiating the locally uniformly paired Fourier series gives (2.5). $\square$

For real $\alpha, \beta > 1$, the first identity, expressed using (2.3), is the nonsingular law in Sun [2, Remark 3.2]. The new use here is at its singular integer parameters and their spectral jets. At positive integers,

$$R_m(x) = -\frac{B_m(\{x\})}{m!} \quad (m \geq 1),$$

up to immaterial point values of the sawtooth for $m = 1$. At negative integers,

$$R_{-k} = \delta_0^{(k)} \quad (k \geq 1). \quad (2.6)$$

The quotient in (2.3) vanishes pointwise at $\alpha = -k$ away from the integers, while the distribution is nonzero. This is the basic reason contact terms must be retained.

3 Canonical finite parts of the Stieltjes functions

The generalized Stieltjes functions are defined for $x > 0$ by

$$\zeta(1-z, x) = -\frac{1}{z} + \sum_{n=0}^{\infty} \gamma_n(x) \frac{z^n}{n!}. \quad (3.1)$$

Thus $\gamma_0(x) = -\psi(x)$. The Hurwitz shift relation gives

$$\gamma_n(x) = \frac{\log^n x}{x} + \gamma_n(1+x). \quad (3.2)$$

Consequently, periodic extension of γ_n is not locally integrable at 0; a convention is necessary.

Definition 3.1 (Canonical periodic finite part). For a smooth periodic test function ϕ , define

$$\langle G_n, \phi \rangle = \int_0^1 \left(\gamma_n(x) \phi(x) - \frac{\log^n x}{x} \phi(0) \right) dx \quad (3.3)$$

$$= \lim_{\epsilon \downarrow 0} \left\{ \int_\epsilon^1 \gamma_n(x) \phi(x) dx + \frac{\log^{n+1} \epsilon}{n+1} \phi(0) \right\}. \quad (3.4)$$

The integrand in (3.3) is integrable: its singular part is $\log^n x [\phi(x) - \phi(0)]/x$, which is $O(|\log x|^n)$. No arbitrary finite multiple of δ_0 is added.

Lemma 3.2 (The distributional Laurent expansion). *Let*

$$Z(s) = \Gamma(1-s)R_{1-s}.$$

It is a meromorphic periodic-distribution family, agreeing with the periodic locally integrable Hurwitz zeta for $\operatorname{Re} s < 1$. Near $z = 0$,

$$Z(1-z) = \frac{1}{z} + \sum_{n=0}^{\infty} G_n \frac{z^n}{n!}. \quad (3.5)$$

Every G_n has mean zero.

Proof. The meromorphic continuation follows from Lemma 2.2. For $\operatorname{Re} z > 0$, split the pairing using $\zeta(1-z, x) = x^{z-1} + \zeta(1-z, 1+x)$. The first term has expansion

$$\int_0^1 x^{z-1} \phi(x) dx = \frac{\phi(0)}{z} + \sum_{n \geq 0} \frac{z^n}{n!} \int_0^1 \frac{\log^n x}{x} [\phi(x) - \phi(0)] dx.$$

The second term contributes $-z^{-1} \int_0^1 \phi$ and the coefficients $\int_0^1 \gamma_n(1+x)\phi(x) dx$. These expansions converge locally near zero after pairing, by the same endpoint domination used above. Combining them proves (3.5) with the exact subtraction in Definition 3.1. Since $\hat{Z}(s)(0) = 0$, every coefficient has mean zero. Equivalently, direct integration over $[1, 2]$ gives $\int_1^2 \zeta(1-z, x) dx = -1/z$, whose regular coefficients vanish. $\square$

Remark 3.3 (Two different residues). The scalar function in (3.1) has residue -1 as a function of z , whereas the periodic-distribution family has residue $\delta_0 - 1$. Analytic continuation across a nonintegrable endpoint adds a point-supported residue. Treating the scalar residue as the whole periodic residue gives wrong convolution constants.

4 All-index convolution closure and exact normal forms

Write $\mathcal{G}(z) = \sum_{n \geq 0} G_n z^n / n!$, holomorphic as a distribution family for $|z| < 1$. By (3.5),

$$1 + z\mathcal{G}(z) = \Gamma(1+z)R_z. \quad (4.1)$$

Let $X = G_0$. For a complete exponential Bell polynomial we use

$$\exp \left(\sum_{j \geq 1} b_j \frac{z^j}{j!} \right) = \sum_{r \geq 0} \mathcal{B}_r(b_1, \dots, b_r) \frac{z^r}{r!}. \quad (4.2)$$

When its arguments are distributions, all products in this definition are convolution products and its scalar unit is 1 .

Theorem 4.1 (One-generator convolution algebra). *For every $n \geq 0$,*

$$(n+1)G_n = \mathcal{B}_{n+1}(b_1, \dots, b_{n+1}), \quad b_1 = X, \quad b_j = (-1)^j (j-1)! \zeta(j) 1 \quad (j \geq 2). \quad (4.3)$$

The span of $1, G_0, G_1, \dots$ is a convolution algebra isomorphic to the polynomial algebra $\mathbb{C}[X]$. These distributions form a basis of that algebra, with the unit listed first.

Proof. For $k \neq 0$, the Fourier transform of (4.1) is

$$1 + z\hat{\mathcal{G}}(z)(k) = \Gamma(1+z)e^{-z\lambda_k} = \exp \left[(-\gamma - \lambda_k)z + \sum_{j \geq 2} \frac{(-1)^j \zeta(j)}{j} z^j \right].$$

The linear coefficient is $\hat{X}(k) = -\gamma - \lambda_k$. Comparing with (4.2) proves (4.3); the zero Fourier coefficient of each distributional expression vanishes. The leading term in G_n is $X^{*(n+1)}/(n+1)$, so the listed distributions span exactly the polynomial algebra generated by X .

There are no polynomial relations among its powers. Indeed, if $P(X) = 0$ distributionally, then $P(-\gamma - \log(2\pi k) - i\pi/2) = 0$ for every positive integer k . These are infinitely many distinct complex numbers, and hence P is the zero polynomial. This also proves the basis assertion. $\square$

This algebraic freeness concerns distributions as functions of the variable x . It is not an independence theorem about their values at a fixed rational point, nor about zeta or Stieltjes constants as numbers.

The first Bell forms are

$$\begin{aligned} 2G_1 &= X^{*2} + \zeta(2)\mathbf{l}, \\ 3G_2 &= X^{*3} + 3\zeta(2)X - 2\zeta(3)\mathbf{l}, \\ 4G_3 &= X^{*4} + 6\zeta(2)X^{*2} - 8\zeta(3)X + [3\zeta(2)^2 + 6\zeta(4)]\mathbf{l}. \end{aligned} \tag{4.4}$$

They already imply many exact integrals. An efficient all-index multiplication rule is supplied by the following Gamma quotient.

Theorem 4.2 (Bivariate multiplication law). *Define, near $(u, v) = (0, 0)$,*

$$\begin{aligned} B(u, v) &= \frac{\Gamma(1+u)\Gamma(1+v)}{\Gamma(1+u+v)} \\ &= \exp \left(\sum_{j \geq 2} \frac{(-1)^j \zeta(j)}{j} [u^j + v^j - (u+v)^j] \right). \end{aligned} \tag{4.5}$$

Then

$$\begin{aligned} uv \mathcal{G}(u) * \mathcal{G}(v) &= [B(u, v) - 1]\mathbf{l} \\ &\quad + B(u, v)(u+v)\mathcal{G}(u+v) - u\mathcal{G}(u) - v\mathcal{G}(v). \end{aligned} \tag{4.6}$$

The right side is divisible by uv as a convergent distribution-valued power series. In particular,

$$G_m * G_n = \frac{m+n+2}{(m+1)(n+1)} G_{m+n+1} + \text{span}_{\mathbb{C}}\{\mathbf{l}, G_0, \dots, G_{m+n}\}, \tag{4.7}$$

and every coefficient is a polynomial over $\mathbb{Q}$ in zeta values at positive integers. Its complete value is obtained by multiplying the coefficient of $u^{m+1}v^{n+1}$ in (4.6) by $m!n!$.

Proof. Convolve (4.1) at parameters u and v and apply Theorem 2.3. The result is

$$[\mathbf{l} + u\mathcal{G}(u)] * [\mathbf{l} + v\mathcal{G}(v)] = B(u, v)[\mathbf{l} + (u+v)\mathcal{G}(u+v)].$$

Subtract the unit and the two linear summands. Both restrictions $u = 0$ and $v = 0$ vanish identically, proving divisibility. Coefficient extraction is legitimate in a sufficiently small bidisc. The largest-index term comes from $(u+v)\mathcal{G}(u+v)$ with the constant coefficient of B ; its coefficient is $\frac{m!n!}{(m+n+1)!} \binom{m+n+2}{m+1}$, which equals the number in (4.7). The exponential expansion of B proves the zeta-polynomial assertion. $\square$

Four useful multiplication rules are

$$G_0 * G_0 = 2G_1 - \zeta(2)\mathbf{l}, \quad (4.8)$$

$$G_0 * G_1 = \frac{3}{2}G_2 - \zeta(2)G_0 + \zeta(3)\mathbf{l}, \quad (4.9)$$

$$G_0 * G_2 = \frac{4}{3}G_3 - 2\zeta(2)G_1 + 2\zeta(3)G_0 - 2\zeta(4)\mathbf{l}, \quad (4.10)$$

$$G_1 * G_1 = G_3 - 2\zeta(2)G_1 + 2\zeta(3)G_0 - \frac{1}{4}\zeta(4)\mathbf{l}. \quad (4.11)$$

The last simplification uses $\zeta(2)^2 = 5\zeta(4)/2$. The verification catalog retains independent formal symbols for zeta values until such a specific relation is explicitly applied.

5 Ordinary integral realizations and explicit Stieltjes identities

The singular support of G_n is contained in $\{0\}$. Thus its convolution with another G_m is smooth away from 0: in a local convolution integral at $0 < x < 1$, the two singular factors occur at the distinct integration points $t = 0$ and $t = x$. We now give a direct formula that also proves this local realization without requiring a product of distributions at the same point.

Theorem 5.1 (Absolutely convergent subtracted integrals). *Put $a_n(t) = \log^n(t)/t$. For $m, n \geq 0$ and $0 < x < 1$, define*

$$\begin{aligned} C_{m,n}(x) = & \int_0^x [\gamma_m(t)\gamma_n(x-t) - \gamma_n(x)a_m(t) - \gamma_m(x)a_n(x-t)] dt \\ & + \int_x^1 \gamma_m(t)\gamma_n(1+x-t) dt \\ & + \gamma_n(x)\frac{\log^{m+1}x}{m+1} + \gamma_m(x)\frac{\log^{n+1}x}{n+1}. \end{aligned} \quad (5.1)$$

*Both displayed integrals are absolutely convergent, and $C_{m,n}(x) = (G_m * G_n)(x)$. In particular, every expression (5.1) has a finite exact reduction using Theorem 4.2.*

Proof. Near $t = 0$, the shift relation (3.2) leaves a possible singularity

$$a_m(t)[\gamma_n(x-t) - \gamma_n(x)] = O(|\log t|^m).$$

The other endpoint is the same argument with m, n interchanged. The second integral is nonsingular. These observations prove absolute convergence, locally uniformly with derivatives in x after using endpoint-fixed local coordinates.

For the normalization, approximate G_m by the cutoff function $\mathbf{1}_{[\epsilon,1)}\gamma_m$ plus $\log^{m+1}\epsilon \delta_0/(m+1)$, and similarly for G_n . On a compact subinterval of $0 < x < 1$, their convolution consists of

$$\int_\epsilon^{x-\epsilon} \gamma_m(t)\gamma_n(x-t) dt + \int_x^1 \gamma_m(t)\gamma_n(1+x-t) dt$$

plus the two logarithmic counterterms multiplying $\gamma_n(x)$ and $\gamma_m(x)$. The product of the two delta counterterms is supported at 0, outside that compact subinterval. Integrating the two subtracted a -terms explicitly and taking the limit gives (5.1). The locally uniform endpoint argument identifies this limit with the restriction of the distributional convolution. $\square$

Corollary 5.2 (Four scalar identity families). *For $0 < x < 1$,*

$$C_{0,0}(x) = 2\gamma_1(x) + \zeta(2), \quad (5.2)$$

$$C_{0,1}(x) = \frac{3}{2}\gamma_2(x) - \zeta(2)\gamma_0(x) - \zeta(3), \quad (5.3)$$

$$C_{0,2}(x) = \frac{4}{3}\gamma_3(x) - 2\zeta(2)\gamma_1(x) + 2\zeta(3)\gamma_0(x) + 2\zeta(4), \quad (5.4)$$

$$C_{1,1}(x) = \gamma_3(x) - 2\zeta(2)\gamma_1(x) + 2\zeta(3)\gamma_0(x) + \frac{1}{4}\zeta(4). \quad (5.5)$$

Proof. Restrict (4.8)–(4.11) to $0 < x < 1$, where G_n agrees with γ_n and $\mathbf{l}$ agrees with the constant -1 . $\square$

5.1 A concrete digamma integral

Since $\gamma_0 = -\psi$, the first case reads

$$\begin{aligned} \int_0^x \left[\psi(t)\psi(x-t) + \psi(x) \left(\frac{1}{t} + \frac{1}{x-t} \right) \right] dt \\ + \int_x^1 \psi(t)\psi(1+x-t) dt - 2\psi(x) \log x = 2\gamma_1(x) + \frac{\pi^2}{6}. \end{aligned} \quad (5.6)$$

The two divergent-looking summands inside the first bracket must be integrated together; their sum is integrable. Equation (5.6) is not an assertion that either unregularized convolution integral exists.

At $x = 1/2$, the Hurwitz identity $\zeta(s, 1/2) = (2^s - 1)\zeta(s)$ implies

$$\gamma_1(1/2) = \gamma_1 - 2\gamma \log 2 - \log^2 2.$$

Thus the left side of (5.6) at $1/2$ equals

$$2\gamma_1 - 4\gamma \log 2 - 2\log^2 2 + \frac{\pi^2}{6}. \quad (5.7)$$

The same method at any rational argument turns the full multiplication table into exact relations among rational-argument Stieltjes functions, with no integer-relation search.

6 Differentiation and the contact-term correction

For $k \geq 1$, the ordinary function $\gamma_n^{(k)}(x)$ has a power-log singularity of order $k+1$. Its canonical Hadamard finite part on $\mathbb{T}$ is defined by

$$\langle F_{n,k}, \phi \rangle = \text{CT}_{\epsilon \downarrow 0} \int_{\epsilon}^1 \gamma_n^{(k)}(x) \phi(x) dx. \quad (6.1)$$

Here CT means the coefficient independent of ϵ after the finite expansion in negative powers of ϵ and nonconstant powers of $\log \epsilon$ is removed. The remaining error tends to zero. This convention uses the coordinate $x \in (0, 1)$ and no finite counterterm. It agrees with Definition 3.1 when $k = 0$.

Lemma 6.1 (Higher Hurwitz poles as distributions). *For each $k \geq 1$, as $z \rightarrow 0$,*

$$Z(1+k-z) = \frac{(-1)^k \delta_0^{(k)}}{k! z} + \sum_{j \geq 0} \frac{(-1)^j z^j}{j!} \text{Fp}_{\mathbb{T}}[\zeta^{(j)}(k+1, x)]. \quad (6.2)$$

The finite parts in this formula have the cutoff convention (6.1).

Proof. Split $\zeta(1+k-z, x) = x^{-k-1+z} + \zeta(1+k-z, 1+x)$. Subtract the Taylor polynomial of ϕ at zero through degree k . Its remainder gives an analytic integral near $z = 0$. The subtracted monomials give terms $\phi^{(r)}(0)/(r!(z+r-k))$, $0 \leq r \leq k$; only $r = k$ has a pole at zero. Since $\langle \delta_0^{(k)}, \phi \rangle = (-1)^k \phi^{(k)}(0)$, its residue is the one stated. Expanding each regular term and each analytic remainder is exactly the constant-term prescription for the cutoff integrals. This proves the entire Laurent expansion. $\square$

Theorem 6.2 (All-index differentiation anomaly). *For $n \geq 0$ and $k \geq 1$,*

$$D^k G_n = F_{n,k} + (-1)^{n+1} n! e_{n+1}(k) \delta_0^{(k)}. \quad (6.3)$$

In particular, there is no anomaly when $n \geq k$.

Proof. The meromorphic distribution family satisfies

$$D^k Z(s) = (-1)^k (s)_k Z(s+k). \quad (6.4)$$

This follows directly from its Fourier coefficients and the Gamma recurrence; it also follows by repeated integration by parts in a left half-plane and meromorphic continuation. Put $s = 1 - z$ and expand

$$(1-z)_k = k! \prod_{r=1}^k (1-z/r) = k! \sum_{i=0}^k (-1)^i e_i(k) z^i.$$

On the left of (6.4), the pole is $\delta_0^{(k)}/z$ and the regular coefficient of z^n is $D^k G_n/n!$. On the right, multiplying the regular part of Lemma 6.1 by the polynomial gives $F_{n,k}/n!$: this is the finite part of the ordinary pointwise parameter-derivative identity. Multiplying its polar part by that same polynomial gives, in addition, the regular coefficient $(-1)^{n+1} e_{n+1}(k) \delta_0^{(k)}$. Comparison proves the formula. $\square$

The first examples are

$$\begin{aligned} DG_0 &= F_{0,1} - \delta'_0, \\ D^2 G_0 &= F_{0,2} - \frac{3}{2} \delta''_0, \\ D^2 G_1 &= F_{1,2} + \frac{1}{2} \delta''_0. \end{aligned} \quad (6.5)$$

The point-supported terms disappear upon restriction to $(0, 1)$, so ordinary differentiation there cannot detect them. But convolving $\delta_0^{(k)}$ with another distribution differentiates that other factor everywhere. Therefore these terms cannot be discarded before convolution, even when the desired final identity is off the singular point.

6.1 Relationship with the manuscript's master identity

The pointwise formula used in the existing derivative chapter is

$$\gamma_n^{(k)}(x) = (-1)^{n+k} k! n! \sum_{i+j=n} e_i(k) \frac{\zeta^{(j)}(k+1, x)}{j!}. \quad (6.6)$$

It follows by expanding $\partial_x^k \zeta(s, x) = (-1)^k (s)_k \zeta(s+k, x)$ at $s = 1$, and is consistent with Coffey's all-order formulas [7]. Theorem 6.2 neither contradicts nor modifies (6.6); it corrects the *additional, unjustified* step of commuting canonical finite part with periodic differentiation. This distinction should be retained in any integration into the manuscript.

6.2 Convolutions of arbitrary differentiated Stieltjes functions

The same argument gives a closed law for all derivative indices, not only the zeroth Stieltjes function. Put $F_{n,0} = G_n$ and define

$$a_{n,k} = (-1)^{n+1} n! e_{n+1}(k) \quad (k \geq 1), \quad a_{n,0} = 0.$$

Thus $F_{n,k} = D^k G_n - a_{n,k} \delta_0^{(k)}$ when $k \geq 1$.

Corollary 6.3 (The full differentiated convolution calculus). *Suppose $n, m, k, l \geq 0$, $r = k + l \geq 1$, and the exact rule from Theorem 4.2 is*

$$G_n * G_m = c_{-1} \mathbf{I} + \sum_{j=0}^{n+m+1} c_j G_j.$$

Then

$$\begin{aligned} F_{n,k} * F_{m,l} &= \sum_{j=0}^{n+m+1} c_j D^r G_j - a_{n,k} D^r G_m - a_{m,l} D^r G_n \\ &\quad + (c_{-1} + a_{n,k} a_{m,l}) \delta_0^{(r)}. \end{aligned} \tag{6.7}$$

Away from the integer point this reduces to

$$\sum_{j=0}^{n+m+1} c_j \gamma_j^{(r)}(x) - a_{n,k} \gamma_m^{(r)}(x) - a_{m,l} \gamma_n^{(r)}(x).$$

In particular, every such convolution has a finite reduction to $\zeta^{(j)}(r+1, x)$ with $0 \leq j \leq n+m+1$.

Proof. Multiply the two identities $F = DG - a\delta$ at their indicated orders, interpreting the zero-order case through $a_{n,0} = 0$. The product of the first terms is $D^r(G_n * G_m)$, and $D^r \mathbf{I} = \delta_0^{(r)}$. Restricting off zero and applying (6.6) proves the last assertion. $\square$

As an explicit higher-index example, the contact term vanishes for $F_{1,1} = DG_1$. Hence

$$\begin{aligned} (F_{1,1} * F_{1,1})(x) &= -2\zeta'''(3, x) - 9\zeta''(3, x) \\ &\quad + [4\zeta(2) - 6]\zeta'(3, x) + [6\zeta(2) + 4\zeta(3)]\zeta(3, x). \end{aligned} \tag{6.8}$$

This follows by twice differentiating (4.11) and using (6.6). It illustrates that the full tower still closes at one spectral integer even when higher Stieltjes indices occur.

7 A two-jet collapse for every polygamma convolution

Let $H_0 = 0$ and $H_k = \sum_{j=1}^k 1/j$ for $k \geq 1$. Define

$$\Psi_k = \text{Fp}_{\mathbb{T}} \psi^{(k)} \quad (k \geq 0)$$

using the same canonical cutoff convention. Since $\gamma_0 = -\psi$, Theorem 6.2 gives

$$\Psi_0 = -G_0, \quad \Psi_k = -D^k G_0 - H_k \delta_0^{(k)} \quad (k \geq 1). \tag{7.1}$$

Theorem 7.1 (Polygamma convolution reduction). *Let $k, l \geq 0$ and $r = k + l \geq 1$. Then*

$$\begin{aligned}\Psi_k * \Psi_l &= 2D^r G_1 + (H_k + H_l)D^r G_0 \\ &\quad + [H_k H_l - \zeta(2)]\delta_0^{(r)}.\end{aligned}\tag{7.2}$$

Consequently, for $0 < x < 1$,

$$(\Psi_k * \Psi_l)(x) = (-1)^{r+1} r! [2\zeta'(r+1, x) + (2H_r - H_k - H_l)\zeta(r+1, x)].\tag{7.3}$$

For $k = l = 0$, the appropriate separate formula is $\Psi_0 * \Psi_0 = 2G_1 - \zeta(2)1$.

Proof. Convolving (7.1) and using that derivatives may be placed on either factor gives

$$\Psi_k * \Psi_l = D^r(G_0 * G_0) + (H_k + H_l)D^r G_0 + H_k H_l \delta_0^{(r)}.$$

The same expression is valid when one index is zero, since its harmonic number vanishes. Apply (4.8) and $D^r 1 = \delta_0^{(r)}$. Restriction to $0 < x < 1$ gives $2\gamma_1^{(r)} + (H_k + H_l)\gamma_0^{(r)}$. Now use (6.6) with $n = 0, 1$:

$$\gamma_1^{(r)} = (-1)^{r+1} r! [\zeta'(r+1, x) + H_r \zeta(r+1, x)], \quad \gamma_0^{(r)} = (-1)^r r! \zeta(r+1, x).$$

This proves (7.3). □

For example,

$$\begin{aligned}(\Psi_0 * \Psi_1)(x) &= 2\zeta'(2, x) + \zeta(2, x), \\ (\Psi_1 * \Psi_1)(x) &= -4\zeta'(3, x) - 2\zeta(3, x), \\ (\Psi_0 * \Psi_2)(x) &= -4\zeta'(3, x) - 3\zeta(3, x).\end{aligned}\tag{7.4}$$

Although the last two convolutions have the same total derivative order, they have different coefficients. Confusing Ψ_k with the k th derivative of Ψ_0 would incorrectly erase this distinction.

7.1 A convergent trigamma identity with every subtraction visible

For $0 < x < 1$, set $h = x/2$ and $f(t) = \psi'(t)$. Then

$$\begin{aligned}\mathcal{T}(x) &= 2 \int_0^h \left[\frac{f(x-t) - f(x) + t f'(x)}{t^2} + \psi'(1+t)f(x-t) \right] dt \\ &\quad + \int_x^1 f(t)f(1+x-t) dt - \frac{2f(x)}{h} - 2f'(x) \log h\end{aligned}\tag{7.5}$$

satisfies

$$\mathcal{T}(x) = -4\zeta'(3, x) - 2\zeta(3, x).\tag{7.6}$$

Indeed, the shift relation $\psi'(t) = t^{-2} + \psi'(1+t)$ and symmetry about $t = x/2$ reduce the first singular convolution integral to twice its left half. Subtracting the constant and linear Taylor terms of $f(x-t)$ leaves a bounded quotient. The constant terms of $\int_\epsilon^h t^{-2} dt$ and $\int_\epsilon^h t^{-1} dt$ are $-1/h$ and $\log h$, respectively. This proves that (7.5) is exactly the canonical finite-part convolution, so Theorem 7.1 applies. Every integral in (7.5) is ordinary and absolutely convergent.

8 Antiderivatives and log-gamma convolution powers

At the integrated end no finite part is needed. Put

$$L = \log(2\pi), \quad Z_j(x) = \zeta^{(j)}(0, x), \quad g(x) = Z_1(x) = \log \Gamma(x) - \frac{L}{2}. \quad (8.1)$$

Each Z_j has mean zero. It is locally a polynomial in $\log x$ plus a smooth function near zero, and hence belongs to every finite $L^p(0, 1)$.

Proposition 8.1 (Centered Stieltjes primitives). *For $n \geq 0$, the unique mean-zero periodic distribution primitive of G_n is the integrable function*

$$A_n(x) = \frac{(-1)^{n+1}}{n+1} \zeta^{(n+1)}(0, x), \quad DA_n = G_n. \quad (8.2)$$

Its ordinary derivative on $(0, 1)$ is $\gamma_n(x)$. The repository's primitive based at 1 is obtained by subtracting the value at 1:

$$\int_1^x \gamma_n(t) dt = A_n(x) - A_n(1).$$

Proof. Differentiate the meromorphic identity $DZ(s) = -sZ(s+1)$ at $s = 0$, using (3.5). Its coefficient of s^{n+1} gives $D\zeta^{(n+1)}(0, x) = (-1)^{n+1}(n+1)G_n$. The mean is zero because $Z(s)$ has mean zero. Finally, a periodic distribution annihilated by D has only a zeroth Fourier coefficient, so imposing mean zero makes the primitive unique. $\square$

Theorem 8.2 (All mixed integrated jets). *Let $d \geq 2$ and $r_1, \dots, r_d \geq 0$ be integers. Write $S = s_1 + \dots + s_d$. Then, as an identity of continuous functions after circular convolution,*

$$Z_{r_1} * \dots * Z_{r_d}(x) = \partial_{s_1}^{r_1} \dots \partial_{s_d}^{r_d} \left[\frac{\prod_{i=1}^d \Gamma(1-s_i)}{\Gamma(d-S)} \zeta(1-d+S, x) \right] \Big|_{s_1=\dots=s_d=0}. \quad (8.3)$$

It holds for $0 < x < 1$ and at the integer point by continuous extension. Thus any mixed convolution of the centered primitives A_n reduces to finitely many jets at the single integer $1-d$.

Proof. Before differentiation, write each Hurwitz factor as $\zeta(s_i, x) = \Gamma(1-s_i)R_{1-s_i}(x)$ and apply Theorem 2.3 repeatedly. For parameters sufficiently close to zero, each factor is holomorphic as an L^2 -valued family: the endpoint singularities are bounded by $x^{-\eta}(1 + |\log x|^M)$ with $2\eta < 1$ on each fixed derivative order and compact parameter set. Convolution of two L^2 functions is continuous, by Cauchy-Schwarz and translation continuity; further convolution with L^1 factors preserves continuity. These same estimates justify differentiating all the convolutions. The right side is continuous at the integer point since $\operatorname{Re}(1-d+S) < 0$ in a small parameter neighborhood. This proves the formula and its endpoint extension. $\square$

The case with every $r_i = 1$ admits a compact all- d formula. Define $H_m^{(j)} = \sum_{a=1}^m a^{-j}$, and, for fixed $d \geq 2$, set

$$b_1 = H_{d-1}, \quad b_j = (j-1)! [H_{d-1}^{(j)} - \zeta(j)] \quad (j \geq 2). \quad (8.4)$$

Theorem 8.3 (All convolution powers of centered log-gamma). *For every integer $d \geq 2$,*

$$g^{*d}(x) = \frac{1}{(d-1)!} \sum_{j=0}^d \binom{d}{j} \mathcal{B}_{d-j}(b_1, \dots, b_{d-j}) \zeta^{(j)}(1-d, x), \quad (8.5)$$

where $\mathcal{B}_0 = 1$ and the b_j are scalar quantities in (8.4).

Proof. In (8.3), differentiate once in each s_i . Only the squarefree coefficient $s_1 \cdots s_d$ matters. Modulo $s_i^2 = 0$, one has

$$\prod_{i=1}^d \Gamma(1 - s_i) = \prod_{i=1}^d (1 + \gamma s_i) = \exp(\gamma S).$$

Consequently the required mixed derivative equals the ordinary d th derivative at $S = 0$ of $e^{\gamma S} \zeta(1 - d + S, x) / \Gamma(d - S)$. Now

$$\log \frac{e^{\gamma S} \Gamma(d)}{\Gamma(d - S)} = H_{d-1} S + \sum_{j \geq 2} \frac{H_{d-1}^{(j)} - \zeta(j)}{j} S^j.$$

This follows directly from $\Gamma(d - S) = \Gamma(1 - S) \prod_{a=1}^{d-1} (a - S)$ and the Taylor series of $\log \Gamma(1 - S)$. Apply (4.2) and the Leibniz rule; $\Gamma(d) = (d - 1)!$ supplies the prefactor. $\square$

The first two nontrivial cases are

$$g * g = \zeta''(-1, x) + 2\zeta'(-1, x) + [2 - \zeta(2)]\zeta(-1, x), \quad (8.6)$$

$$\begin{aligned} g^{*3} &= \frac{1}{2}\zeta'''(-2, x) + \frac{9}{4}\zeta''(-2, x) + \left[\frac{21}{4} - \frac{3}{2}\zeta(2)\right]\zeta'(-2, x) \\ &\quad + \left[\frac{45}{8} - \frac{9}{4}\zeta(2) - \zeta(3)\right]\zeta(-2, x). \end{aligned} \quad (8.7)$$

The first left side is explicitly

$$\int_0^x g(t)g(x - t) dt + \int_x^1 g(t)g(1 + x - t) dt.$$

For uncentered $f = \log \Gamma$, mean-zero convolution annihilates all mixed constant/nonconstant terms, so

$$f^{*d}(x) = \left(\frac{L}{2}\right)^d + g^{*d}(x). \quad (8.8)$$

Remark 8.4 (Convolution powers are not pointwise moments). The value $g^{*3}(0)$ is a two-dimensional circular-convolution integral. It is not $\int_0^1 g(t)^3 dt$. Thus (8.7) does not eliminate the Tornheim derivative coordinates of the ordinary cubic log-gamma moment discussed in the repository. Likewise, the mixed reflected pointwise moments in `07-reflected-moments.tex` are different objects. This distinction prevents a spurious claim of resolving that separate reduction problem.

9 Reflection, polylogarithmic order derivatives, and shifted correlations

For a distribution T , let $\tilde{T}(x) = T(-x)$; there is no complex conjugation in this definition. Reflection changes the sign of its Fourier index. This produces a different, polylogarithmic, multiplication law from the unreflected Hurwitz semigroup.

Theorem 9.1 (Reflected spectral master identity). *For $\operatorname{Re} \alpha, \operatorname{Re} \beta > 1$ and $0 \leq x \leq 1$,*

$$\begin{aligned} (R_\alpha * \tilde{R}_\beta)(x) &= (2\pi)^{-\alpha-\beta} [e^{\pi i(\beta-\alpha)/2} \operatorname{Li}_{\alpha+\beta}(e^{2\pi i x}) \\ &\quad + e^{-\pi i(\beta-\alpha)/2} \operatorname{Li}_{\alpha+\beta}(e^{-2\pi i x})]. \end{aligned} \quad (9.1)$$

The right side, interpreted by its periodic Fourier coefficients, continues to all $(\alpha, \beta) \in \mathbb{C}^2$ as a distribution-valued identity. Arbitrary parameter derivatives of this equation are valid.

Proof. For $k > 0$, the Fourier multiplier on the left is

$$(2\pi ik)^{-\alpha}(-2\pi ik)^{-\beta} = (2\pi k)^{-\alpha-\beta} e^{\pi i(\beta-\alpha)/2}.$$

For $k < 0$ the phase is reversed. Summing the two absolutely convergent series proves the ordinary identity. For general parameters, the coefficients and their parameter derivatives are polynomially bounded on compact sets, so the paired Fourier series define the entire continuation and justify differentiation. $\square$

We write

$$\text{Li}_s^{[j]}(z) = \frac{\partial^j}{\partial s^j} \text{Li}_s(z)$$

for an *order* derivative. This must not be confused with z -differentiation, which instead lowers the order through $z\partial_z \text{Li}_s(z) = \text{Li}_{s-1}(z)$.

Corollary 9.2 (Full shifted log-gamma correlation). *Put $L = \log(2\pi)$ and $A = \gamma + L$. For $0 \leq x \leq 1$, with endpoint values understood by continuity,*

$$\begin{aligned} \int_0^1 \log \Gamma(t) \log \Gamma(\{t - x\}) dt &= \frac{L^2}{4} \\ &+ \frac{1}{2\pi^2} \text{Re} \left[\text{Li}_2^{[2]}(e^{2\pi ix}) - 2A \text{Li}_2^{[1]}(e^{2\pi ix}) + \left(A^2 + \frac{\pi^2}{4} \right) \text{Li}_2(e^{2\pi ix}) \right]. \end{aligned} \quad (9.2)$$

Every integral here is an ordinary absolutely convergent integral.

Proof. Differentiating the Hurwitz Fourier coefficient at $s = 0$ gives, for $k \neq 0$,

$$\widehat{g}(k) = \frac{\gamma + \lambda_k}{2\pi ik}.$$

Since g is real-valued, its correlation has coefficient $\widehat{g}(k)\widehat{g}(-k) = |\widehat{g}(k)|^2$. For $k > 0$ this is

$$\frac{(A + \log k)^2 + \pi^2/4}{4\pi^2 k^2}.$$

The coefficients are absolutely summable. Combining the positive and negative indices and using $\text{Li}_s^{[j]}(z) = \sum_{k \geq 1} (-\log k)^j z^k / k^s$ near $s = 2$ proves the formula for centered g . Restoring the mean gives $L^2/4$. The convolution of two L^2 functions is continuous; alternatively, the displayed absolutely convergent Fourier series is uniform in x . At $x = 0$, the integrand has only an integrable squared logarithm. $\square$

At $x = 0$, this recovers the classical second moment studied by Espinosa and Moll [5] and already present in the manuscript:

$$\begin{aligned} \int_0^1 \log^2 \Gamma(t) dt &= \frac{\gamma^2}{12} + \frac{\pi^2}{48} + \frac{\gamma L}{6} + \frac{L^2}{3} \\ &- \frac{\gamma + L}{\pi^2} \zeta'(2) + \frac{\zeta''(2)}{2\pi^2}. \end{aligned} \quad (9.3)$$

This specialization is a consistency check, not a new evaluation claim. The full shift retains the polylogarithmic phase.

At the half shift, put $\ell = \log 2$. Using $\text{Li}_s(-1) = (2^{1-s} - 1)\zeta(s)$ in (9.2) yields

$$\begin{aligned} \int_0^1 \log \Gamma(t) \log \Gamma(\{t - \tfrac{1}{2}\}) dt &= \frac{L^2}{4} - \frac{\zeta''(2)}{4\pi^2} + \frac{A - \ell}{2\pi^2} \zeta'(2) \\ &+ \frac{\zeta(2)}{4\pi^2} \left[\ell^2 + 2A\ell - A^2 - \frac{\pi^2}{4} \right]. \end{aligned} \quad (9.4)$$

Other rational shifts give finite Hurwitz-jet grids, as follows.

Proposition 9.3 (Cyclotomic jet conversion). *Let $q \geq 1$, $p \in \mathbb{Z}$, and $N \geq 2$ be integers. For $r \geq 0$,*

$$\mathrm{Li}_N^{[r]}(e^{2\pi ip/q}) = q^{-N} \sum_{a=1}^q e^{2\pi ipa/q} \sum_{j=0}^r \binom{r}{j} (-\log q)^{r-j} \zeta^{(j)}(N, a/q). \quad (9.5)$$

In particular, every rational-shift correlation in Corollary 9.2 reduces to finitely many jets at $s = 2$.

Proof. For $\mathrm{Re} s > 1$, split the defining Dirichlet series for the polylogarithm into residue classes modulo q :

$$\mathrm{Li}_s(e^{2\pi ip/q}) = q^{-s} \sum_{a=1}^q e^{2\pi ipa/q} \zeta(s, a/q).$$

Absolute locally uniform convergence permits r derivatives in s . The Leibniz rule gives (9.5). Its underlying distribution and polylogarithm identities are classical [3, 4]; here it is the explicit conversion rule for the new convolution calculations. $\square$

10 Audit findings and proposed manuscript integration

No incorrect central theorem was established in the inspected integration and differentiation chapters. The following distinctions and localized corrections should accompany any integration of this continuation.

The singular normalization must be stated. Equations (5.2) and (7.3) require the canonical finite parts in Definitions 3.1 and (6.1). For example, the prescription $D^k \Psi_0$ is a different extension from Ψ_k when $k \geq 1$. Omitting the anomaly would produce $-4\zeta'(3, x) - 6\zeta(3, x)$ instead of (1.3) for the trigamma convolution. This is a correction to a tempting extension of the existing calculus, not a claim that the manuscript prints that false trigamma formula.

A local integration-by-parts clarification. The proof of the Hurwitz moment transform in 07-integration.tex, at label `stieltjes:thm:transform`, treats the lower endpoint through the factor x^m . For $m \geq 1$ its vanishing is appropriate. For the base case $m = 0$, the conclusion $I_0(s) = 0$ instead follows because the primitive $\zeta(s - 1, x)$ has *equal* endpoint limits when $\mathrm{Re} s < 1$; those limits are generally nonzero. The theorem is correct. The proposed clarification is to state the cancellation separately rather than allow the vanishing-endpoint explanation to cover $m = 0$.

A PSLQ wording refinement. The early negative-search note at `integral:neg:psim2` says that basis atoms must be $\mathbb{Q}$ -independent. Complete arithmetic independence is generally unknown and is not an operational prerequisite for a valid exact relation. A more precise instruction is to remove *known* dependencies, require a nonzero target coefficient, and independently prove any proposed relation. The later discovery chapter already expresses this caution more accurately. This is an editorial refinement, not a rejected mathematical identity.

Do not reopen settled items or conflate integral types. At the pinned snapshot, the surviving-programme chapter marks S_4 proved and the cubic log-gamma moment represented through Tornheim coordinates. The present convolution powers do not resolve its further

reduction to a smaller algebra. The finite-part closure also proves no independence among numerical Stieltjes values at a fixed argument.

Placement. A suitable new thematic report is `Analysis/Polylogarithms/docs/reports/stieltjes-convolution/`. The integration chapter can cite Sections 8–9; the derivative chapter can cite Sections 6–7. The full finite-part and Fourier conventions should be retained in a standalone subsection rather than scattered among scalar tables. No remote repository files were changed by this delivery.

11 Verification and reproducibility

The analytic proofs above are independent of numerical experiments. The accompanying program performs two deliberately different kinds of checks.

11.1 Finite exact algebra

The exact suite uses rational arithmetic and formal symbols z_j for $\zeta(j)$. It compares an exponential-series recurrence with an independent complete Bell-polynomial recurrence; compares the product of two one-variable normal forms with a separately assembled bivariate Gamma quotient; checks the leading coefficient for all $0 \leq m, n \leq 4$; checks the four printed low-index laws, applying the even-zeta relation only where needed; compares elementary symmetric coefficients with unsigned Stirling numbers through $k = 12$; directly differentiates the singular term $\log^n(x)/x$ through $k = 6, n = 6$; and checks the squarefree-derivative coefficients of log-gamma convolution powers through $d = 7$.

There are 229 exact assertions. The emitted `identity_catalog.json` contains the 25 finite Stieltjes product rules and the six log-gamma convolution-power coefficient lists. The finite checks certify those polynomial identities relative to their stated input expansions. They are not a proof-assistant verification of the infinite-dimensional analytic results.

11.2 Independent numerical diagnostics

The numerical suite compares ordinary quadrature of the endpoint-subtracted digamma, higher Stieltjes, and trigamma integrals with independent Stieltjes and Hurwitz-zeta routines at $x = 0.27, 0.5, 0.81$. It also checks the ordinary log-gamma convolution, a complex-order semigroup instance, shifted correlations, cyclotomic order jets, a Fourier-mode contact term, and the half-point Stieltjes specialization. There are 31 numerical checks, including seven cross-checks of the Taylor evaluator used for γ_1 . The machine-readable output records the actual precision and residual for every check. These are floating-point diagnostics, not interval enclosures or equality certificates.

Two numerical details are important. Near a subtracted endpoint, the code evaluates removable quotients by Taylor polynomials rather than subtracting nearly equal floating-point numbers. For order derivatives of polylogarithms near an integer order, a fixed small differentiation step is used when necessary; a naively tiny default step can amplify internal cancellation. Exact roots such as -1 are supplied exactly, not as rounded complex exponentials. This is a numerical implementation issue, not a change to the analytic formulas.

11.3 Rebuilding

The package uses Python with `mpmath` and `sympy`, and a standard $\text{\LaTeX}$ installation. The verification code makes no network requests. Run

```
python verification/verify.py --exact-only
python verification/verify.py --numeric-only --dps 40
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The README gives the tested dependency versions and the contents of every auxiliary artifact. Test output is written beside the verification script; rerun on a working copy when preserving original diagnostics.

12 Further research questions

The present work proves exact closure for a specific convolution geometry. Several next questions have clear mathematical targets rather than merely larger numerical searches.

12.1 Twists, characters, and alternate endpoint normalizations

For a nonintegral twist θ , the multipliers $(2\pi i(k + \theta))^{-\alpha}$ give a twisted spectral calculus with no vanishing frequency $k + \theta$. Determine the analogue of the finite-part Stieltjes normal form and identify precisely which Dirichlet or Lerch jets replace the ordinary zeta coefficients. A new result should include its singular normalization, not merely a nonsingular convolution identity. A useful first target is an explicit all-index multiplication table for the half-twist, including its point-supported counterterms.

12.2 Affine arguments and the boundary of depth-one reduction

The complete collapse in Theorem 8.2 relies on matching Fourier indices. If a factor has a different affine frequency, or a pointwise product occurs before convolution, multiple independent summation indices appear. Classify the integer-affine configurations that still reduce to a single Hurwitz jet. For the remaining configurations, identify the minimal natural multiple-zeta, colored polylogarithm, or Tornheim family. This would sharpen the distinction between convolution powers and the ordinary cubic moment rather than blur it.

12.3 Collision limits and renormalization compatibility

The scalar formulas in Section 5 are stated for $0 < x < 1$. When $x \rightarrow 0$, the two singularities collide. Determine a complete local expansion that simultaneously tracks logarithmic terms, power terms, and contact distributions, and compare taking the finite part in x with first taking finite parts in the integration variable. A precise target is an explicit commutator formula for these two operations at arbitrary Stieltjes indices. The two-jet polygamma collapse gives a tractable test family.

12.4 Cyclotomic integration certificates

For a fixed denominator q , combine (9.5) with the repository’s character-coordinate and distribution certificates. The objective is a division-controlled exact certificate translating a

rational-shift correlation into the chosen character-jet basis, with all conductor factors visible. Any rank statement should be about its specified presentation; arithmetic independence after evaluation is a separate problem.

12.5 Negative-integer jets and arithmetic reductions

Formula (8.5) organizes a large family of ordinary integrals at one negative spectral integer. Which rational-argument combinations simplify further under reflection, distribution, and Dirichlet functional equations? The cases $d = 3, 4$ at denominators 3, 4, 5 are concrete starting points. A meaningful reduction claim must specify the comparison algebra; replacing a Hurwitz jet by an equivalent Dirichlet derivative is a coordinate change unless that target algebra is explicitly part of the objective.

12.6 Minimal formalization interfaces

A proof-assistant development can begin with finite Bell recurrences, the Gamma-quotient coefficient identity, and the elementary-symmetric contact coefficients. Its analytic interface then needs Fourier uniqueness for periodic distributions, convolution multiplication, and the endpoint subtraction lemma. Establishing the distributional Laurent expansion once should make the remaining spectral-jet proofs short. None of those analytic formalization steps is claimed complete in this delivery.

13 Conclusion

A single Fourier normalization links four apparently different layers: ordinary log-gamma convolution integrals, generalized Stieltjes finite parts, polygamma convolutions, and polylogarithmic correlations. The nonsingular Hurwitz semigroup is the classical organizing identity. At its singular parameter, the residue $\delta_0 - 1$ creates a closed one-generator Stieltjes convolution algebra. Its higher poles determine the exact derivative contact terms. Retaining those terms yields the correct two-jet polygamma reduction; moving in the opposite direction produces all log-gamma convolution powers and their mixed primitive analogues. These are explicit identities with proved domains and normalizations, rather than conjectures supported only by small numerical residuals.

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